

HOMEWORK 1 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Question Q1.

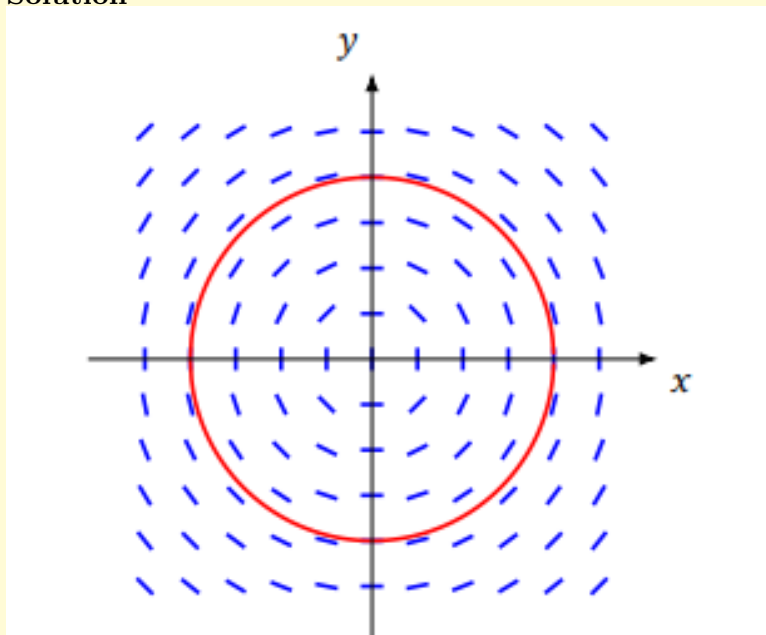
a) Construct a slope field of the differential equation

$$y' = -\frac{x}{y}$$

for $y = -3, -2, -1, 0, 1, 2, 3$ and your own choice of x -values.

b) From your slope field, **sketch** one integral curve.

Solution



The solution to part (a) is the blue ticks only. The solution to part (b) is the red circle.

Question Q2. Find the solution to the differential equation

$$y' = \frac{1}{x^2 + x}; y(0) = 1.$$

In other words, integrate the function y' .

Solution

We can integrate $y' = \frac{1}{x^2 + x}$ by partial fractions, but the initial condition at $x = 0$ is problematic.

$$\frac{1}{x^2 + x} = \frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}.$$

Hence

$$y(x) = \int \left(\frac{1}{x} - \frac{1}{x+1} \right) dx = \ln|x| - \ln|x+1| + C = \ln \left| \frac{x}{x+1} \right| + C, \quad (x \neq 0, -1).$$

The differential equation is **not defined at** $x = 0$ (the right-hand side blows up), and

$$\lim_{x \rightarrow 0} (\ln|x| - \ln|x+1| + C) = -\infty$$

for any constant C . Therefore it is ****impossible**** to choose C so that $y(0) = 1$.

The general solution (on any interval avoiding $x = 0$ and $x = -1$) is

$$y(x) = \ln \left| \frac{x}{x+1} \right| + C$$

$\lim_{x \rightarrow 0} y(x) = -\infty$. Thus **the IVP has no solution**.

0.1. Question Q3.

a) Which of the following is the general solution to the differential equation

$$y' - 2y + e^x = 0.$$

- i) $y = ce^{2x} + x$
- ii) $y = ce^{2x} + e^{-x}$
- iii) $y = ce^x + e^{-x}$
- iv) $y = e^x - ce^{2x}$

b) From your answer in (a), solve the IVP

$$y' - 2y + e^x = 0, \quad y(0) = 2.$$

Solution

I'll show the derivative and the substitution for each option to determine which satisfies the differential equation.

- i) $y = ce^{2x} + x \Rightarrow y' = 2ce^{2x} + 1$
Substitute:

$$y' - 2y + e^x = (2ce^{2x} + 1) - 2(ce^{2x} + x) + e^x = 1 - 2x + e^x \quad \text{not a solution.}$$

- ii) $y = ce^{2x} + e^{-x} \Rightarrow y' = 2ce^{2x} - e^{-x}$
Substitute:

$$y' - 2y + e^x = (2ce^{2x} - e^{-x}) - 2(ce^{2x} + e^{-x}) + e^x = -3e^{-x} + e^x, \quad \text{not a solution}$$

iii) $y = ce^x + e^{-x} \Rightarrow y' = ce^x - e^{-x}$

Substitute:

$$y' - 2y + e^x = (ce^x - e^{-x}) - 2(ce^x + e^{-x}) + e^x = (1 - c)e^x - 3e^{-x}, \quad \text{not a solution}$$

iv) $y = e^x - ce^{2x} \Rightarrow y' = e^x - 2ce^{2x}$

Substitute:

$$y' - 2y + e^x = (e^x - 2ce^{2x}) - 2(e^x - ce^{2x}) + e^x = e^x - 2ce^{2x} - 2e^x + 2ce^{2x} + e^x = 0.$$

Therefore (iv) is a valid general solution ($y = e^x + Ce^{2x}$ with $C = -c$).

Part (b): Solve the IVP $y' - 2y + e^x = 0$, $y(0) = 2$

Use the valid form $y(x) = e^x - ce^{2x}$. Apply the initial condition $x = 0$:

$$y(0) = e^0 - ce^0 = 1 - c = 2 \quad \Rightarrow \quad c = -1.$$

Thus the particular solution is

$$y(x) = e^x + e^{2x}$$

The correct option is (iv). The solution of the IVP is $y(x) = e^x + e^{2x}$.